

Definition 1 The *unit sphere* consists of all points at unit distance from the origin.

In $\mathbb{R}^n$ with an inner product $\langle \cdot, \cdot \rangle$, the unit sphere is denoted by S^{n-1} or $\mathcal{S}^{n-1}$.

Example 2 The complex inner product in $\mathbb{C} = \mathbb{R}^2$ is

$$\langle z, w \rangle := z^* \cdot w := \bar{z}^T \cdot w = (x - i \cdot y) \cdot (u + i \cdot v) = (x \cdot u + y \cdot v) + i \cdot (x \cdot v - y \cdot u).$$

In $\mathbb{C}^2 = \mathbb{C} \times \mathbb{C} = \mathbb{R}^2 \times \mathbb{R}^2 = \mathbb{R}^4$ the complex inner product is

$$\begin{aligned} \langle \mathbf{z}, \mathbf{w} \rangle &:= \mathbf{z}^* \cdot \mathbf{w} := \bar{\mathbf{z}}^T \cdot \mathbf{w} = \begin{pmatrix} \bar{z}_1 & \bar{z}_2 \end{pmatrix} \cdot \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \bar{z}_1 \cdot w_1 + \bar{z}_2 \cdot w_2 \\ &= (x_1 - i \cdot y_1) \cdot (u_1 + i \cdot v_1) + (x_2 - i \cdot y_2) \cdot (u_2 + i \cdot v_2) \\ &= (x_1 \cdot u_1 + y_1 \cdot v_1) + i \cdot (x_1 \cdot v_1 - y_1 \cdot u_1) + (x_2 \cdot u_2 + y_2 \cdot v_2) + i \cdot (x_2 \cdot v_2 - y_2 \cdot u_2). \end{aligned}$$

The squared magnitude of a vector in $\mathbb{C}^2$ matches the squared magnitude of a vector in $\mathbb{R}^4$:

$$\begin{aligned} \|\mathbf{z}\|^2 &= \left\| \begin{pmatrix} z_1 & z_2 \end{pmatrix} \right\|^2 = \langle \mathbf{z}, \mathbf{z} \rangle = \mathbf{z}^* \cdot \mathbf{z} := \bar{\mathbf{z}}^T \cdot \mathbf{z} = \begin{pmatrix} \bar{z}_1 & \bar{z}_2 \end{pmatrix} \cdot \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \bar{z}_1 \cdot z_1 + \bar{z}_2 \cdot z_2 \\ &= (x_1 - i \cdot y_1) \cdot (x_1 + i \cdot y_1) + (x_2 - i \cdot y_2) \cdot (x_2 + i \cdot y_2) \\ &= (x_1 \cdot x_1 + y_1 \cdot y_1) + i \cdot (x_1 \cdot y_1 - y_1 \cdot x_1) + (x_2 \cdot x_2 + y_2 \cdot y_2) + i \cdot (x_2 \cdot y_2 - y_2 \cdot x_2) \\ &= x_1^2 + y_1^2 + x_2^2 + y_2^2 = |z_1|^2 + |z_2|^2. \end{aligned}$$

The unit sphere in $\mathbb{C}^2$ is thus

$$\mathcal{S}^3 = \{\mathbf{z} \in \mathbb{C}^2 : \|\mathbf{z}\| = 1\}.$$

Definition 3 On the unit sphere $\mathcal{S}^3 \subset \mathbb{C}^2$ define a relation $R \subset \mathcal{S}^3 \times \mathcal{S}^3$ by $(\mathbf{z}, \mathbf{w}) \in R$, which is abbreviated by $\mathbf{z}R\mathbf{w}$, if and only if there is a real number γ such that $\mathbf{z} = e^{i\gamma} \cdot \mathbf{w}$.

Theorem 4 The relation R in Definition 3 is an equivalence relation.

Proof. Reflexivity: $\mathbf{z}R\mathbf{z}$ for every $\mathbf{z} \in \mathcal{S}^3$, because $\mathbf{z} = 1 \cdot \mathbf{z} = e^{i \cdot 0} \cdot \mathbf{z}$.

Symmetry: If $\mathbf{z}R\mathbf{w}$, then $\mathbf{z} = e^{i\gamma} \cdot \mathbf{w}$, whence $e^{-i\gamma} \cdot \mathbf{z} = \mathbf{w}$, which states that $\mathbf{w}R\mathbf{z}$.

Transitivity: If $\mathbf{z}R\mathbf{w}$ and $\mathbf{w}R\mathbf{t}$, then $\mathbf{z} = e^{i\gamma} \cdot \mathbf{w}$ and $\mathbf{w} = e^{i\beta} \cdot \mathbf{t}$, whence $\mathbf{z} = e^{i(\gamma+\beta)} \cdot \mathbf{t}$, which states that $\mathbf{z}R\mathbf{t}$.

Definition 5 If (ψ_0, ψ_1) is an ordered orthonormal basis of $\mathbb{C}^2$, then $\psi \in \mathcal{S}^3 \subset \mathbb{C}^2$, equivalently, $\|\psi\| = 1$, if and only if there are complex coefficients c_0 and c_1 such that

$$\psi = c_0 \cdot \psi_0 + c_1 \cdot \psi_1, \quad |c_0|^2 + |c_1|^2 = 1.$$

Denote the polar coordinates of $(|c_0|, |c_1|)$ by $(1, \theta/2)$. Then $0 \leq \theta \leq \pi$, because $|c_0|, |c_1| \geq 0$. Thus, $|c_0| = \cos(\theta/2)$ and $|c_1| = \sin(\theta/2)$. Hence, $c_0 = e^{i\gamma} \cdot \cos(\theta/2)$ and $c_1 = e^{i\beta} \cdot \sin(\theta/2)$:

$$\begin{aligned} \psi &= c_0 \cdot \psi_0 + c_1 \cdot \psi_1 = e^{i\gamma} \cdot \cos(\theta/2) \cdot \psi_0 + e^{i\beta} \cdot \sin(\theta/2) \cdot \psi_1 \\ &= e^{i\gamma} \cdot [\cos(\theta/2) \cdot \psi_0 + e^{i\phi} \cdot \sin(\theta/2) \cdot \psi_1]. \end{aligned}$$

Theorem 6 *The set $\mathcal{S}^3/R$ of equivalence classes for the relation R in Definition 3 is isomorphic to the unit sphere $\mathcal{S}^2 \subset \mathbb{R}^3$ with spherical coordinates, longitude ϕ and co-latitude θ .*

Proof. Denote the equivalence class of ψ by $[\psi]$:

$$[\psi] = [\cos(\theta/2) \cdot \psi_0 + e^{i\cdot\phi} \cdot \sin(\theta/2) \cdot \psi_1],$$

with $0 \leq \theta \leq \pi$ and $-\pi < \phi \leq \pi$ but $\phi = 0$ if $\theta \in \{0, \pi\}$ unique to ψ . Thus, the isomorphism is

$$F : \mathcal{S}^3/R \rightarrow \mathcal{S}^3, \quad [\cos(\theta/2) \cdot \psi_0 + e^{i\cdot\phi} \cdot \sin(\theta/2) \cdot \psi_1] \mapsto \begin{pmatrix} \sin(\theta) \cdot \cos(\phi) \\ \sin(\theta) \cdot \sin(\phi) \\ \cos(\theta) \end{pmatrix}$$

Example 7

$$\begin{aligned} [\psi_0] &= | +z \rangle = [\cos(0/2) \cdot \psi_0 + e^{i\cdot 0} \cdot \sin(0/2) \cdot \psi_1], & \theta &= 0, & F(| +z \rangle) &= (0, 0, 1)^T = \mathbf{e}_z; \\ [\psi_1] &= | -z \rangle = [\cos(\pi/2) \cdot \psi_0 + e^{i\cdot 0} \cdot \sin(\pi/2) \cdot \psi_1], & \theta &= \pi, & F(| -z \rangle) &= (0, 0, -1)^T; \\ | +x \rangle &= [\cos(\pi/4) \cdot \psi_0 + e^{i\cdot 0} \cdot \sin(\pi/4) \cdot \psi_1], & \theta &= \pi/2, & F(| +x \rangle) &= (1, 0, 0)^T = \mathbf{e}_x; \\ | +y \rangle &= [\cos(\pi/4) \cdot \psi_0 + e^{i\cdot\pi/2} \cdot \sin(\pi/4) \cdot \psi_1], & \phi &= \pi/2, & F(| +y \rangle) &= (0, 1, 0)^T = \mathbf{e}_y; \end{aligned}$$

Exercise 1 Calculate $F(| -x \rangle)$ and $F(| -y \rangle)$.

A rotation of the Bloch Sphere in the Euclidean space corresponds to a unitary transformation of states in $\mathbb{H}_1 = \mathbb{C}^2$. The proof takes up a chapter in reference [1, Ch. 4].

References

- [1] Johannes A. Buchmann. *Introduction to quantum algorithms*, volume 64 of *Pure and Applied Undergraduate Texts*. American Mathematical Society, Providence, RI, 2024.